DEPI Graduation Project Dax Measures

1. Availability

DAX Measure

```
Availability =
DIVIDE(
   SUM(Fact_Production_Log[Operating_Time_mins]),
   SUM(Fact_Production_Log[Planned_Time_mins])
)
```

Description
Measures the percentage of planned production time during which the machine was actually operating. A higher Availability value indicates less downtime and better equipment utilization.

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2. Performance

DAX Measure

```
Performance =
DIVIDE(
   SUM(Fact_Production_Log[Actual_Production]),
   DIVIDE(
      SUM(Fact_Production_Log[Operating_Time_mins]),
      AVERAGE(Fact_Production_Log[Ideal_Cycle_Time_mins_per_piece])
   )
)
```

Description
Evaluates how efficiently the machine operates compared to its ideal production speed. It compares actual production output with the maximum achievable output during operating time.

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3. Quality

DAX Measure

```
Quality =
DIVIDE(
   SUM(Fact_Production_Log[Total_Pieces]) -
   SUM(Fact_Production_Log[Defective_Pieces]),
   SUM(Fact_Production_Log[Total_Pieces])
)
```

Description
Measures the proportion of defect-free products produced during manufacturing. Higher values indicate better product quality and fewer defective items.

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4. Overall Equipment Effectiveness (OEE)

DAX Measure

```
OEE =
[Availability] * [Performance] * [Quality]
```

Description
A key manufacturing KPI that combines Availability, Performance, and Quality into a single metric to evaluate overall equipment effectiveness. World-class manufacturing operations typically achieve an OEE of 85% or higher.

Downtime Metrics

5. Downtime Percentage

DAX Measure

```
Downtime % =
DIVIDE(
   SUM(Fact_Production_Log[Downtime_Minutes]),
   SUM(Fact_Production_Log[Planned_Time_mins])
) * 100
```

Description
Represents the percentage of planned production time lost due to equipment downtime. It helps identify the impact of interruptions on production efficiency.

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6. Failure Frequency

DAX Measure

```
Failure Frequency =
CALCULATE(
    COUNTROWS(Fact_Production_Log),
    Fact_Production_Log[Failure_Flag] = 1
)
```

Description
Counts the total number of recorded failure events. This metric helps monitor how frequently equipment failures occur within the production process.

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7. Most Frequent Failure

DAX Measure

```
Most Frequent Failure =
CALCULATE(
    FIRSTNONBLANK(Dim_Failure_Types[Failure_Type], 1),
    TOPN(
        1,
        SUMMARIZE(
            FILTER(
                Fact_Production_Log,
                Fact_Production_Log[Failure_Flag] = 1
            ),
            Dim_Failure_Types[Failure_Type],
            "cnt", COUNTROWS(Fact_Production_Log)
        ),
        [cnt], DESC
    )
)
```

Description
Identifies the failure type that occurs most frequently. This measure helps maintenance teams prioritize root-cause analysis and corrective actions.

Reliability Metrics

8. Mean Time Between Failures (MTBF)

DAX Measure

```
MTBF (hrs) =
DIVIDE(
    SUM(Fact_Production_Log[Operating_Time_mins]) / 60,
    CALCULATE(
        COUNTROWS(Fact_Production_Log),
        Fact_Production_Log[Failure_Flag] = 1
    )
)
```

Description
Measures the average operating time between consecutive failures. Higher MTBF values indicate greater equipment reliability and stability.

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9. Mean Time To Repair (MTTR)

DAX Measure

```
MTTR (mins) =
CALCULATE(
    AVERAGE(Fact_Maintenance_Events[Repair_Duration_Mins]),
    Fact_Maintenance_Events[Failure_ID] <> "F000"
)
```

Description
Measures the average time required to repair equipment after a failure occurs. Lower MTTR values indicate faster maintenance response and recovery.

Quality Metrics

10. Scrap Rate

DAX Measure

```
Scrap Rate % =
DIVIDE(
    SUM(Fact_Production_Log[Defective_Pieces]),
    SUM(Fact_Production_Log[Total_Pieces])
) * 100
```

Description
Represents the percentage of defective products relative to total production. Lower scrap rates indicate better manufacturing quality and reduced waste.

Environmental & Energy Metrics

11. Thermal Stress Index

DAX Measure

```
Thermal Stress Index =
AVERAGEX(
    Fact_Production_Log,
    Fact_Production_Log[Process_Temperature_K] -
    Fact_Production_Log[Air_Temperature_K]
)
```

Description
Measures the average temperature difference between the manufacturing process and the surrounding environment. Higher values may indicate increased thermal stress and a greater likelihood of equipment failures.

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12. Energy per Piece

DAX Measure

```
Energy per Piece =
DIVIDE(
    SUM(Fact_Production_Log[Energy_Consumption_kWh]),
    SUM(Fact_Production_Log[Total_Pieces])
)
```

Description
Calculates the average energy consumed to produce a single unit. This metric is used to evaluate energy efficiency and identify opportunities for reducing energy costs.